



Haritha Vijayan

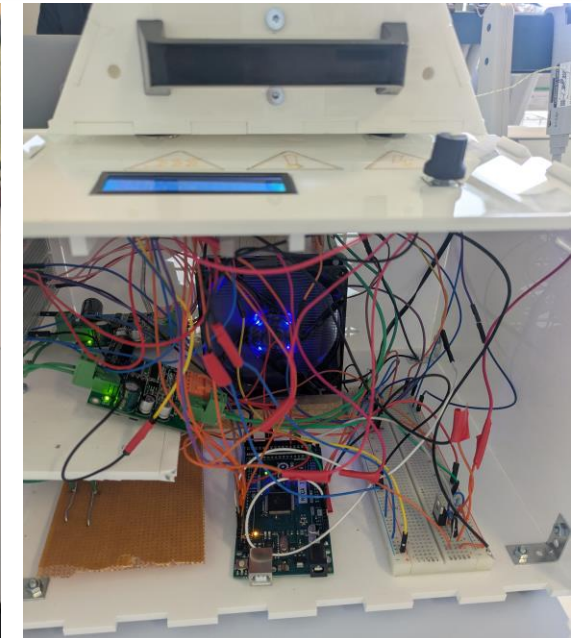
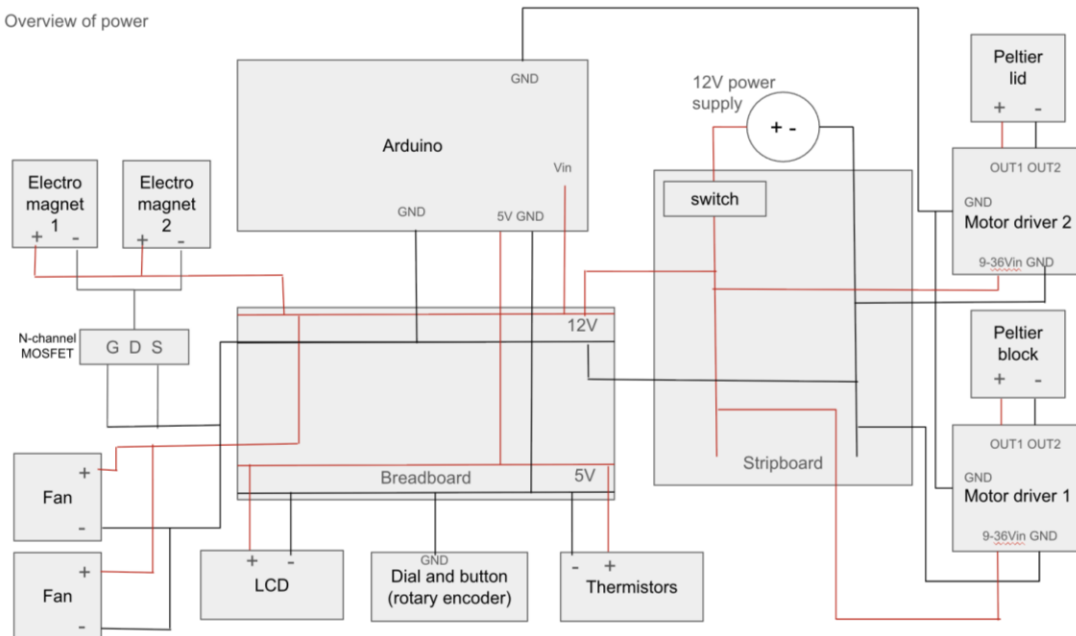
Biomedical Engineering Portfolio

- **PCR machine – design and build group project**
- **Woman's Health in Biomedical Engineering - device design and presentation**
- **CAD – personal development of skills in Fusion 360**

PCR Machine

Task: Design and build a prototype PCR machine which is safe to use, gives a live temperature readout, allows adjustment of the temperature of primer annealing step, costs less than £350, and 'stretch goals' including pre-programmed cycles, password protection, and automatically locking-lid

Overview of power



Action: Contributed design ideas and stretch goals as part of the team, basic calculations and choice of components for a target ramp rate of 1°C/s , designed and built the circuit, programmed the user interface, purchased components and kept track of spending, troubleshooted problems with the electronics.

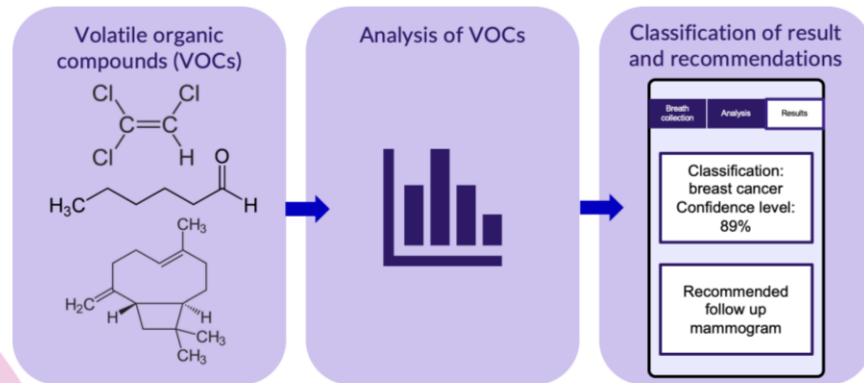
Result: The machine was able to cycle between temperatures but did not reach the 90°C required for DNA replication. All other goals and stretch goals were achieved. Technical specifications include: 3 x 2ml PCR tubes capacity, temperature range $21 - 91.5^{\circ}\text{C}$, temperature accuracy 2.1°C , ramp rate 0.124°C/s .

Women's Health in Biomedical Engineering

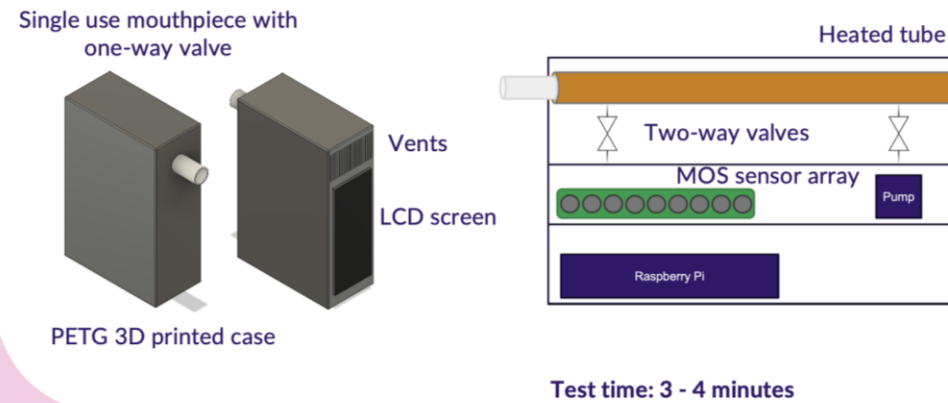
Task: Design an innovation which uses biomedical engineering to improve women's health and pitch it to a venture capitalist for a (hypothetical) £1 million to conduct research and development

Breath Test

Low cost, non-invasive breath test device for use in a primary care setting as an additional screening tool to mammograms



Device



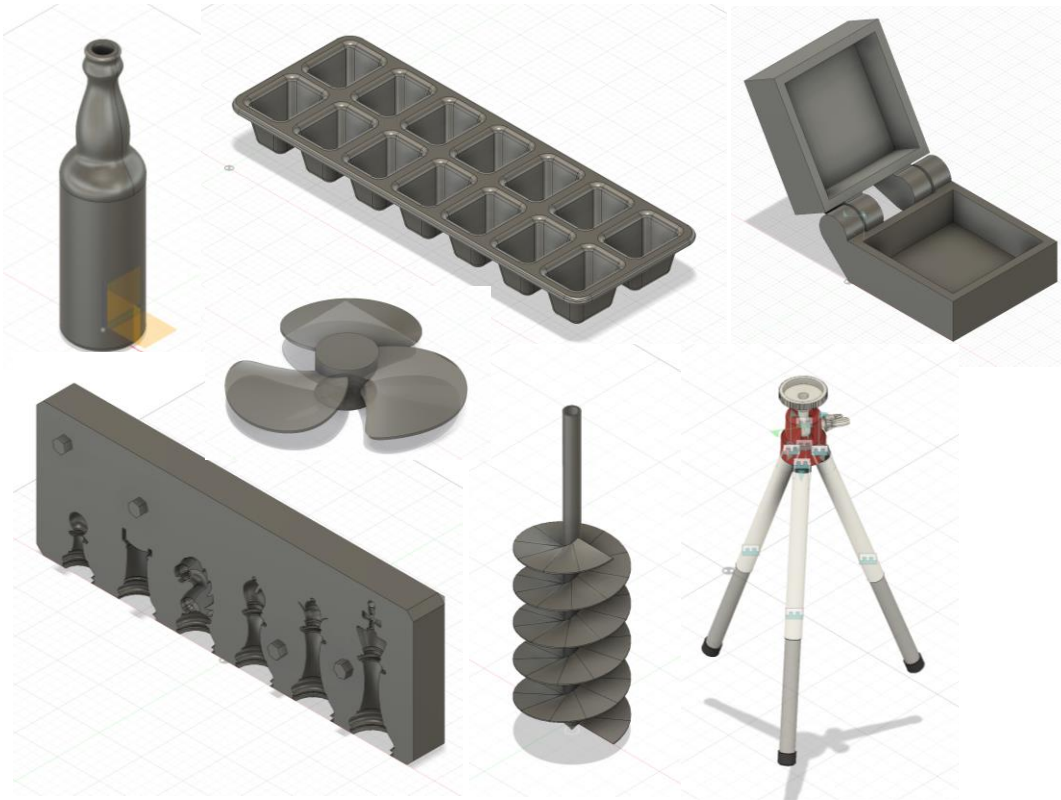
Action: Spoke to family members and learnt about breast cancer diagnosis to find a clinical need. Analysed research papers to create a scientifically based device. Centred the device design and testing around inclusivity. Considered costs, regulations, and the team necessary to bring the device to market. Created and presented a pitch to lecturers and classmates.

Result: A breath test device which uses machine learning to classify presence of breast cancer based on the volatile organic compounds found in the breath. Designed to be low cost, non-invasive, and used in a GP setting. Feedback from the lecturers included good understanding of the technical theory, awareness of limitations, and clear presentation of the clinical need. My presentation was professional and easy to follow. Voted as the 'best idea' by the class.

CAD – Fusion 360

Task: Personal development of CAD skills with a focus on form modelling and 3D printing

Practice designs from the beginner's tutorial



Action: Followed a beginner's tutorial, and practiced applying the techniques that I learnt by creating models.

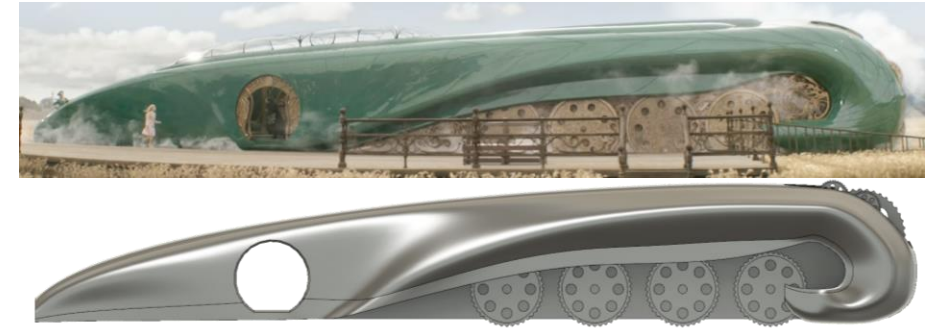
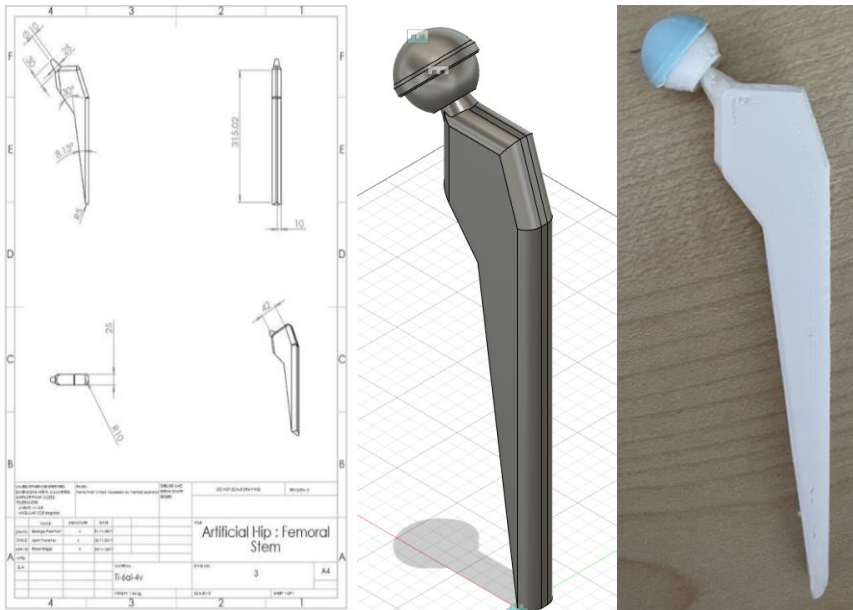
Results: Learnt how to use the following:

- New solid modelling functions such as spline, revolve, sweep, shell, and offset face.
- One-part and two-part moulds including draft angle and registration marks.
- Tolerances for 3D printing moving parts such as hinges.
- Sheet metal, component assemblies, form modelling, and surface modelling.

CAD – Fusion 360

Action: Designed the hip implant from a technical drawing and 3D printed it. Combined form and solid modelling to create a train from a reference image by watching tutorials and iterating through designs until the desired shape was achieved. I used form modelling techniques such as modify, bevel, and subdivide.

Hip implant technical drawing, 3D model, and print



Reference image and model of train

Results: The three pieces of the hip implant fit together well when it was printed, and it matched closely to the technical drawing, but there were a few discrepancies due to missing information in the drawing. The shape of the train resembles the reference image, although there are a lot more details which I could have included.